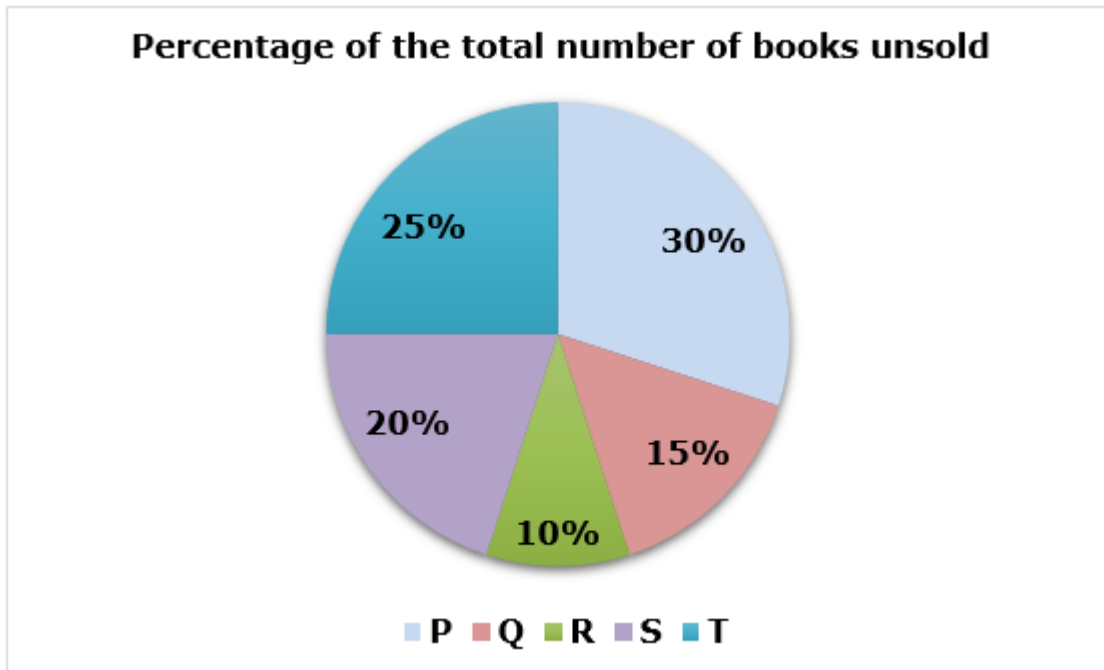


1. Questions

Read the following information carefully and answer the questions based on it.

The pie chart given below shows the percentage distribution of the total number of books unsold by five book stores P, Q, R, S and T.



Note:

The total number of books published = the number of books sold + the number of books unsold

The total number of books sold = the number of books sold through online mode + the number of books sold through offline mode

The ratio of the number of books sold and unsold by book store P, Q, R, S and T is 3:1, 9:1, 4:1, 3:1 and 4:3 respectively.

The number of books sold through online mode by book store P, Q, R, S and T is 600, 1080, 300, 440, 240 respectively.

The difference between the number of books sold through offline mode by book store P and Q is 60. ($P < Q$)

Out of the total number of books published by book store P and Q together, 55.55% are fiction books. If the ratio of the number of fiction and non-fiction books published by book store P is 9:7, then find the number of non-fiction books published by book store Q.

- 640
- 810
- 700
- Cannot be determined
- None of these

2. Questions

In book store R, if the price of each book is Rs. 100, which is sold at a profit of $[X]\%$, then the overall profit earned on selling all books together will be Rs. 12000. Find the value of $[X/5]$.

- a. 10
- b. 15
- c. 20
- d. 25
- e. None of these

3. Questions

In another book store Z, if the total number of books published is 200 less than that of book store T and the number of books unsold in book store Z is equal to the number of books sold through online mode in the same book store. If the ratio of the number of books sold through online and offline mode in book store Z is 2:1, then find the total number of books sold (online + offline) by book store Z.

- a. 240
- b. 400
- c. 180
- d. 300
- e. None of these

4. Questions

A: The difference between the total number of books sold and unsold by book store S.

B: The percentage of the number of books sold through offline mode by book store T.

Find the value of $[A/3B]$.

- a. 12
- b. 5
- c. 4
- d. 15
- e. None of these

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5. Questions

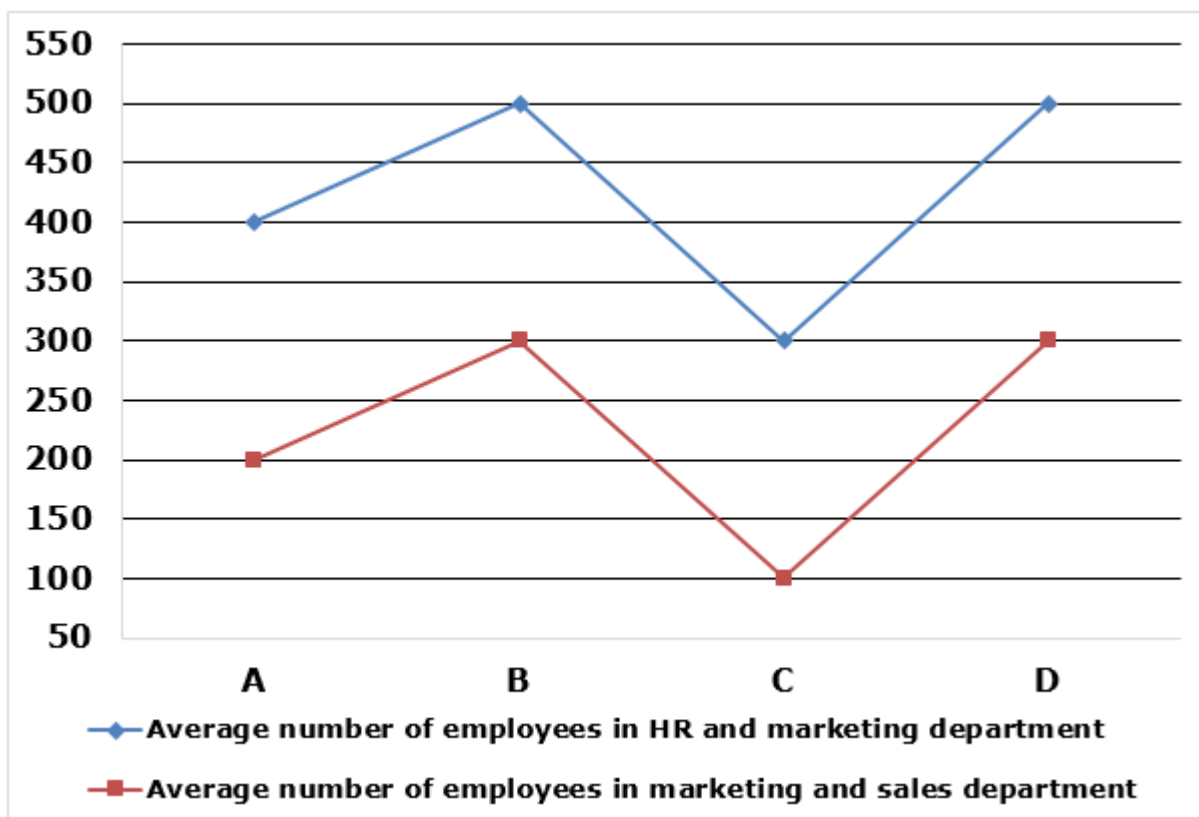
Out of the total number of books sold by book store T, 25% are sold to females. If the number of books sold to males through offline mode is 100, then find the ratio of the number of books sold to males and females through online mode by book store T.

- a. 3:2
- b. 5:3
- c. 2:1
- d. Cannot be determined
- e. None of these

6. Questions

Read the following information carefully and answer the questions based on it.

The line graph given below shows the average number of employees in HR and marketing departments of four companies A, B, C and D. It also shows the average number of employees in marketing and sales departments of each company.



Note:

The total number employees = the number of employees in HR department + the number of employees in marketing department + the number of employees in sales department

The ratio of the number of employees in HR department of company A, B, C and D is 10:15:9:12.

The total number of employees in sales department of all four companies together is 700.

In company A, if the number of male employees in marketing department is 180, which is 9 times of the number of female employees in sales department, then find the ratio of the number of female employees in marketing and male employees in sales department of company A.

- a. 3:2

- b. 2:1
- c. 4:3
- d. 5:3
- e. None of these

7. Questions

In company B, if the number of graduate employees in marketing department is 150, which is 30% of the number of graduate employees in HR department, then find the difference between the number of post-graduate employees in HR and marketing department of company B.(Only graduates and post graduates are work in all the departments)

- a. 175
- b. 225
- c. 200
- d. 150
- e. None of these

8. Questions

Which of the following is/are definitely true?

I: The ratio of the number of employees in HR and sales department of company C is 3:2.

II: The difference between the number of employees in marketing department of company D and C is 200.

III: The total number of employees in company B is 1500.

- a. Only I
- b. Both I and II
- c. Both II and III
- d. All I, II and III
- e. None is true

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9. Questions

The ratio of the number of employees working in HR and marketing departments of another company F is 4:3. The number of employees working in sales department of company F is half of company D. If the total number of employees working in company F is 800, then find the number of employees working in marketing department of company F.

- a. 400
- b. 100

- c. 300
- d. 200
- e. None of these

10. Questions

X = The difference between the number of employees working in marketing and sales departments of company B.

Y = The number of employees working in sales department of company A.

- a. $X > Y$
- b. $X < Y$
- c. $X \geq Y$
- d. $X \leq Y$
- e. $X = Y$

11. Questions

The two series given below. Both series has one missing term. Both series follows different pattern.

Series I: 11, 15, 8, 18, __[P]__, 21

Series II: 5, 7, 17, 55, 225, __[Q]__

Which of the following is/are definitely true?

I: Both [P] and [Q] are exactly divisible by 5.

II: [Q] is a four-digit natural number.

III: [P] is an odd composite number.

- a. Only II
- b. Both I and II
- c. Only III
- d. All I, II and III
- e. None is true

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12. Questions

A series is given below, which has one wrong term.

Series I: 8, 32, 4, 12, 2, 8

Find the correct term that will replace the wrong term of the given series.

- a. 4^2

- b. 3^3
- c. 2^4
- d. both a) and c)
- e. None of these

13. Questions

A series is given below, which has one missing term.

Series: 3, 19, 55, 119, 219, __[?]__

What should come in place of the question mark?

- a. 441
- b. 363
- c. 500
- d. 616
- e. None of these

14. Questions

Two series are given below. Both series have one wrong term. Each series follows the same pattern.

Series I: 40, 48, 64, 92, 160, 288

Series II: 20, 28, 44, 76, 140, 260

Find the difference between the wrong terms of both series.

- a. 168
- b. 220
- c. 138
- d. 224
- e. None of these

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15. Questions

A series is given below, which has one missing term.

Series I: 12, 13, 5, 32, __[?]__, 93

What should come in place of the question mark?

- a. 32
- b. 28

- c. -32
- d. -28
- e. None of these

16. Questions

Read the following information carefully and answer the questions based on it.

There are certain number of people who have two different types of mobiles i.e., android and iPhone from two different cities P and Q. Each person has at least one type of mobile.

City P: The ratio of the number of people who have only android and the number of people who have only iPhone is 2:1. The number of people who have both android and iPhone is 100. The total number of people who have iPhone is 40% less than the total number of people who have android.

City Q: The number of people who have both android and iPhone is 60. The number of people who have only iPhone is $\frac{3}{4}$ th of the number of people who have only android. The total number of people who have iPhone is 20% less than the total number of people who have android.

Note:

The total number of people who have mobiles = the number of people who have only android + the number of people who have only iPhone + the number of people who have both android and iPhone

The total number of people who have android = the number of people who have only android + the number of people who have both android and iPhone

The total number of people who have iPhone = the number of people who have only iPhone + the number of people who have both android and iPhone

In city P, if the total number of people who have 5G android is 50% more than the total number of people who have 5G iPhone. If the total number of people who have 4G android is 150 more than the total number of people who have 4G iPhone, then find the total number of people who have 5G mobiles (android + iPhone).

- a. 150
- b. 250
- c. 100
- d. 200
- e. None of these

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17. Questions

In city Q, if the total number of males who have android is 175, which is 15 more than the total number of males who have iPhone, then find the ratio of the total number of females who have android and iPhone.

- a. 25:16

- b. 17:8
- c. 32:21
- d. Cannot be determined
- e. None of these

18. Questions

The total number of people who have mobiles in another city R is 140 less than city Q. In city R, if the number of people who have both android and iPhone in city R is 80 and the ratio of the number of people of who have only android and only iPhone is 8:5, the find the total number of people who have android in city R.

- a. 320
- b. 180
- c. 400
- d. 240
- e. None of these

19. Questions

If the total number of people who have symbian mobiles in city P is 72% of the total number of people who have android phone in the same city, then find the difference between the total number of people who have iPhone and symbian mobiles in city P.

- a. 75
- b. 60
- c. 44
- d. 32
- e. None of these

20. Questions

In city Q, if the number of people who have both android and iPhone is increased by $[X]$, while the number of people who have only android and only iPhone remained same, then the total number of people who have mobiles in city Q will be 500. Find the value of $[\sqrt{(X-4)}]$.

- a. 4
- b. 9
- c. 5
- d. 6
- e. None of these

Explanations:

1. Questions

Let the total number of books unsold by all five book stores together be 100k.

So, the number of books unsold by book store P = 30% of 100k = 30k

And, the total number of books sold by book store P = $\{(30k/1) \times 3\} = 90k$

The number of books sold through online mode by book store P = 600

The number of books sold through offline mode by book store P = (90k-600)

Now,

The number of books unsold by book store Q = 15% of 100k = 15k

And, the total number of books sold by book store Q = $\{(15k/1) \times 9\} = 135k$

The number of books sold through online mode by book store Q = 1080

The number of books sold through offline mode by book store Q = (135k-1080)

Now,

$$(135k-1080) - (90k-600) = 60$$

$$135k - 1140 = 90k - 600$$

$$45k = 540$$

$$k = 12$$

Therefore, the total number of books unsold by all five book stores together = 100k = $(100 \times 12) = 1200$

Book Store P

The number of books unsold = 30% of 1200 = 360

The number of books sold = $\{(360/1) \times 3\} = 1080$

The total number of books published = $(1080+360) = 1440$

The number of books sold through online mode = 600

The number of books sold through offline mode = $(1080-600) = 480$

Book Store Q

The number of books unsold = 15% of 1200 = 180

The number of books sold = $\{(180/1) \times 9\} = 1620$

The total number of books published = $(1620+180) = 1800$

The number of books sold through online mode = 1080

The number of books sold through offline mode = $(1620-1080) = 540$

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Book Store R

The number of books unsold = 10% of 1200 = 120

The number of books sold = $\{(120/1) \times 4\} = 480$

The total number of books published = $(480+120) = 600$

The number of books sold through online mode = 300

The number of books sold through offline mode = $(480-300) = 180$

Book Store S

The number of books unsold = 20% of 1200 = 240

The number of books sold = $\{(240/1) \times 3\} = 720$

The total number of books published = $(720+240) = 960$

The number of books sold through online mode = 440

The number of books sold through offline mode = $(720-440) = 280$

Book Store T

The number of books unsold = 25% of 1200 = 300

The number of books sold = $\{(300/3) \times 4\} = 400$

The total number of books published = $(400+300) = 700$

The number of books sold through online mode = 240

The number of books sold through offline mode = $(400-240) = 160$

Book Store	The total number of books published	The number of books sold	The number of books unsold	The number of books sold through online mode	The number of books sold through offline mode
P	1440	1080	360	600	480
Q	1800	1620	180	1080	540
R	600	480	120	300	180
S	960	720	240	440	280
T	700	400	300	240	160
Total	5500	4300	1200	2660	1640

Answer: B

Now, according to the question,

The total number of books published by book store P and Q together = $(1440+1800) = 3240$

The total number of fiction books published by book store P and Q together = $5/9$ of 3240 = 1800(**Note:**

$$55.55\% = 5/9)$$

The total number of non-fiction books published by book store P and Q together = $4/9$ of $3240 = 1440$

(Note: $44.44\% = 4/9$)

The total number of books published by book store P = 1440

The number of non-fiction books published by book store P = $7/16$ of $1440 = 630$

Therefore, the number of non-fiction books published by book store Q = $(1440-630) = 810$.

Hence, the correct answer is option B.

2. Questions

Let the total number of books unsold by all five book stores together be $100k$.

So, the number of books unsold by book store P = 30% of $100k = 30k$

And, the total number of books sold by book store P = $\{(30k/1) \times 3\} = 90k$

The number of books sold through online mode by book store P = 600

The number of books sold through offline mode by book store P = $(90k-600)$

Now,

The number of books unsold by book store Q = 15% of $100k = 15k$

And, the total number of books sold by book store Q = $\{(15k/1) \times 9\} = 135k$

The number of books sold through online mode by book store Q = 1080

The number of books sold through offline mode by book store Q = $(135k-1080)$

Now,

$$(135k-1080) - (90k-600) = 60$$

$$135k - 1140 = 90k - 600$$

$$45k = 540$$

$$k = 12$$

Therefore, the total number of books unsold by all five book stores together = $100k = (100 \times 12) = 1200$

Book Store P

The number of books unsold = 30% of $1200 = 360$

The number of books sold = $\{(360/1) \times 3\} = 1080$

The total number of books published = $(1080+360) = 1440$

The number of books sold through online mode = 600

The number of books sold through offline mode = $(1080-600) = 480$

Book Store Q

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The number of books unsold = 15% of 1200 = 180

The number of books sold = $\{(180/1) \times 9\} = 1620$

The total number of books published = $(1620+180) = 1800$

The number of books sold through online mode = 1080

The number of books sold through offline mode = $(1620-1080) = 540$

Book Store R

The number of books unsold = 10% of 1200 = 120

The number of books sold = $\{(120/1) \times 4\} = 480$

The total number of books published = $(480+120) = 600$

The number of books sold through online mode = 300

The number of books sold through offline mode = $(480-300) = 180$

Book Store S

The number of books unsold = 20% of 1200 = 240

The number of books sold = $\{(240/1) \times 3\} = 720$

The total number of books published = $(720+240) = 960$

The number of books sold through online mode = 440

The number of books sold through offline mode = $(720-440) = 280$

Book Store T

The number of books unsold = 25% of 1200 = 300

The number of books sold = $\{(300/3) \times 4\} = 400$

The total number of books published = $(400+300) = 700$

The number of books sold through online mode = 240

The number of books sold through offline mode = $(400-240) = 160$

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Book Store	The total number of books published	The number of books sold	The number of books unsold	The number of books sold through online mode	The number of books sold through offline mode
P	1440	1080	360	600	480
Q	1800	1620	180	1080	540
R	600	480	120	300	180
S	960	720	240	440	280
T	700	400	300	240	160
Total	5500	4300	1200	2660	1640

Answer: A

Book Store R

The total number of books published = 600

So, the overall cost price of all books published = $(600 \times 100) = \text{Rs. } 60000$

The selling price of each book sold = $\{100 + (X\% \text{ of } 100)\} = \text{Rs. } (100 + X)$

The total number of books sold = 480

So, the overall selling price of all books sold = $\{480 \times (100 + X)\} = \text{Rs. } (48000 + 480X)$

Now,

$$60000 + 12000 = 48000 + 480X$$

$$72000 = 48000 + 480X$$

$$480X = 24000$$

$$X = 50$$

Therefore, the value of $X = (50/5) = 10$.

Hence, the correct answer is option A.

3. Questions

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Let the total number of books unsold by all five book stores together be 100k.

So, the number of books unsold by book store P = 30% of 100k = 30k

And, the total number of books sold by book store P = $\{(30k/1) \times 3\} = 90k$

The number of books sold through online mode by book store P = 600

The number of books sold through offline mode by book store P = $(90k - 600)$

Now,

The number of books unsold by book store Q = 15% of 100k = 15k

And, the total number of books sold by book store Q = $\{(15k/1) \times 9\} = 135k$

The number of books sold through online mode by book store Q = 1080

The number of books sold through offline mode by book store Q = $(135k - 1080)$

Now,

$$(135k - 1080) - (90k - 600) = 60$$

$$135k - 1140 = 90k - 600$$

$$45k = 540$$

$$k = 12$$

Therefore, the total number of books unsold by all five book stores together = $100k = (100 \times 12) = 1200$

Book Store P

The number of books unsold = 30% of 1200 = 360

The number of books sold = $\{(360/1) \times 3\} = 1080$

The total number of books published = $(1080 + 360) = 1440$

The number of books sold through online mode = 600

The number of books sold through offline mode = $(1080 - 600) = 480$

Book Store Q

The number of books unsold = 15% of 1200 = 180

The number of books sold = $\{(180/1) \times 9\} = 1620$

The total number of books published = $(1620 + 180) = 1800$

The number of books sold through online mode = 1080

The number of books sold through offline mode = $(1620 - 1080) = 540$

Book Store R

The number of books unsold = 10% of 1200 = 120

The number of books sold = $\{(120/1) \times 4\} = 480$

The total number of books published = $(480 + 120) = 600$

The number of books sold through online mode = 300

The number of books sold through offline mode = $(480 - 300) = 180$

Book Store S

The number of books unsold = 20% of 1200 = 240

The number of books sold = $\{(240/1) \times 3\} = 720$

The total number of books published = $(720 + 240) = 960$

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The number of books sold through online mode = 440

The number of books sold through offline mode = $(720-440) = 280$

Book Store T

The number of books unsold = 25% of 1200 = 300

The number of books sold = $\{(300/3) \times 4\} = 400$

The total number of books published = $(400+300) = 700$

The number of books sold through online mode = 240

The number of books sold through offline mode = $(400-240) = 160$

Book Store	The total number of books published	The number of books sold	The number of books unsold	The number of books sold through online mode	The number of books sold through offline mode
P	1440	1080	360	600	480
Q	1800	1620	180	1080	540
R	600	480	120	300	180
S	960	720	240	440	280
T	700	400	300	240	160
Total	5500	4300	1200	2660	1640

Answer: D

Book store Z

The total number of books sold = $(700-200) = 500$

Let the number of books unsold be n .

So, the total number of books sold = $(500-n)$

The number of books sold through online mode = n

The number of books sold through offline mode = $\{(500-n)-n\} = (500-2n)$

Now,

$$n/(500-2n) = 2/1$$

$$n = 1000 - 4n$$

$$5n = 1000$$

$$n = 200$$

Therefore, the total number of books sold (online + offline) by book store S = $(500-n) = (500-200) = 300$.

Hence, the correct answer is option D.

4. Questions

Let the total number of books unsold by all five book stores together be 100k.

So, the number of books unsold by book store P = 30% of 100k = 30k

And, the total number of books sold by book store P = $\{(30k/1) \times 3\} = 90k$

The number of books sold through online mode by book store P = 600

The number of books sold through offline mode by book store P = $(90k - 600)$

Now,

The number of books unsold by book store Q = 15% of 100k = 15k

And, the total number of books sold by book store Q = $\{(15k/1) \times 9\} = 135k$

The number of books sold through online mode by book store Q = 1080

The number of books sold through offline mode by book store Q = $(135k - 1080)$

Now,

$$(135k - 1080) - (90k - 600) = 60$$

$$135k - 1140 = 90k - 600$$

$$45k = 540$$

$$k = 12$$

Therefore, the total number of books unsold by all five book stores together = $100k = (100 \times 12) = 1200$

Book Store P

The number of books unsold = 30% of 1200 = 360

The number of books sold = $\{(360/1) \times 3\} = 1080$

The total number of books published = $(1080 + 360) = 1440$

The number of books sold through online mode = 600

The number of books sold through offline mode = $(1080 - 600) = 480$

Book Store Q

The number of books unsold = 15% of 1200 = 180

The number of books sold = $\{(180/1) \times 9\} = 1620$

The total number of books published = $(1620 + 180) = 1800$

The number of books sold through online mode = 1080

The number of books sold through offline mode = $(1620 - 1080) = 540$

Book Store R

The number of books unsold = 10% of 1200 = 120

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The number of books sold = $\{(120/1) \times 4\} = 480$

The total number of books published = $(480+120) = 600$

The number of books sold through online mode = 300

The number of books sold through offline mode = $(480-300) = 180$

Book Store S

The number of books unsold = 20% of 1200 = 240

The number of books sold = $\{(240/1) \times 3\} = 720$

The total number of books published = $(720+240) = 960$

The number of books sold through online mode = 440

The number of books sold through offline mode = $(720-440) = 280$

Book Store T

The number of books unsold = 25% of 1200 = 300

The number of books sold = $\{(300/3) \times 4\} = 400$

The total number of books published = $(400+300) = 700$

The number of books sold through online mode = 240

The number of books sold through offline mode = $(400-240) = 160$

Book Store	The total number of books published	The number of books sold	The number of books unsold	The number of books sold through online mode	The number of books sold through offline mode
P	1440	1080	360	600	480
Q	1800	1620	180	1080	540
R	600	480	120	300	180
S	960	720	240	440	280
T	700	400	300	240	160
Total	5500	4300	1200	2660	1640

Answer: C

A = The total number of books sold by book store S = 720

The total number of books unsold by book store S = 240

So, the required difference = $(720-240) = 480$

A = 480.

B = The total number of books sold by book store T = 400

The number of books sold through offline mode by book store T = 160

So, the required percentage = $\{(160/400) \times 100\} \% = 40\%$

B = 40

Therefore, the value of $= (A/3B) = \{480/(3 \times 40)\} = (480/120) = 4$.

Hence, the correct answer is option C.

5. Questions

Let the total number of books unsold by all five book stores together be 100k.

So, the number of books unsold by book store P = 30% of 100k = 30k

And, the total number of books sold by book store P = $\{(30k/1) \times 3\} = 90k$

The number of books sold through online mode by book store P = 600

The number of books sold through offline mode by book store P = (90k-600)

Now,

The number of books unsold by book store Q = 15% of 100k = 15k

And, the total number of books sold by book store Q = $\{(15k/1) \times 9\} = 135k$

The number of books sold through online mode by book store Q = 1080

The number of books sold through offline mode by book store Q = (135k-1080)

Now,

$$(135k-1080) - (90k-600) = 60$$

$$135k - 1140 = 90k - 600$$

$$45k = 540$$

$$k = 12$$

Therefore, the total number of books unsold by all five book stores together = 100k = $(100 \times 12) = 1200$

Book Store P

The number of books unsold = 30% of 1200 = 360

The number of books sold = $\{(360/1) \times 3\} = 1080$

The total number of books published = $(1080+360) = 1440$

The number of books sold through online mode = 600

The number of books sold through offline mode = $(1080-600) = 480$

Book Store Q

The number of books unsold = 15% of 1200 = 180

The number of books sold = $\{(180/1) \times 9\} = 1620$

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The total number of books published = $(1620+180) = 1800$

The number of books sold through online mode = 1080

The number of books sold through offline mode = $(1620-1080) = 540$

Book Store R

The number of books unsold = 10% of 1200 = 120

The number of books sold = $\{(120/1) \times 4\} = 480$

The total number of books published = $(480+120) = 600$

The number of books sold through online mode = 300

The number of books sold through offline mode = $(480-300) = 180$

Book Store S

The number of books unsold = 20% of 1200 = 240

The number of books sold = $\{(240/1) \times 3\} = 720$

The total number of books published = $(720+240) = 960$

The number of books sold through online mode = 440

The number of books sold through offline mode = $(720-440) = 280$

Book Store T

The number of books unsold = 25% of 1200 = 300

The number of books sold = $\{(300/3) \times 4\} = 400$

The total number of books published = $(400+300) = 700$

The number of books sold through online mode = 240

The number of books sold through offline mode = $(400-240) = 160$

Book Store	The total number of books published	The number of books sold	The number of books unsold	The number of books sold through online mode	The number of books sold through offline mode
P	1440	1080	360	600	480
Q	1800	1620	180	1080	540
R	600	480	120	300	180
S	960	720	240	440	280
T	700	400	300	240	160
Total	5500	4300	1200	2660	1640

Answer: E

Book Store T

The total number of books sold = 400

The total number of books sold to females = 25% of 400 = 100

The total number of books sold to males = 75% of 400 = 300

The total number of books sold through offline mode = 160

The number of books sold to males through offline mode = 100

The number of books sold to females through offline mode = $(160-100) = 60$

The number of books sold to males through online mode = $(300-100) = 200$

The number of books sold to females through online mode = $(100-60) = 40$

Therefore, the required ratio = $200:40 = 5:1$.

Hence, the correct answer is option E.

6. Questions

Let the number of employees in marketing department of company A, B, C and D be $10n$, $15n$, $9n$ and $12n$ respectively.

Company A

The total number of employees in HR and marketing departments = $(400 \times 2) = 800$

The number of employees in HR department = $10n$

So, the number of employees in marketing department = $(800-10n)$

The total number of employees in marketing and sales departments = $(200 \times 2) = 400$

So, the number of employees in sales department of company = $\{400-(800-10n)\} = (10n-400)$

Company B

The total number of employees in HR and marketing departments = $(500 \times 2) = 1000$

The number of employees in HR department = $15n$

So, the number of employees in marketing department = $(1000-15n)$

The total number of employees in marketing and sales departments = $(300 \times 2) = 600$

So, the number of employees in sales department of company = $\{600-(1000-15n)\} = (15n-400)$

Company C

The total number of employees in HR and marketing departments = $(300 \times 2) = 600$

The number of employees in HR department = $9n$

So, the number of employees in marketing department = $(600-9n)$

The total number of employees in marketing and sales departments = $(100 \times 2) = 200$

So, the number of employees in sales department of company = $\{200-(600-9n)\} = (9n-400)$

Company D

The total number of employees in HR and marketing departments = $(500 \times 2) = 1000$

The number of employees in HR department = $12n$

So, the number of employees in marketing department = $(1000-12n)$

The total number of employees in marketing and sales departments = $(300 \times 2) = 600$

So, the number of employees in sales department of company = $\{600-(1000-12n)\} = (12n-400)$

Now, the total number of employees in sales department of all four companies together =
 $\{(10n-400)+(15n-400)+(9n-400)+(12n-400)\} = (46n-1600)$

So,

$$46n - 1600 = 700$$

$$46n = 2300$$

$$n = 50$$

Company A

The number of employees in HR department = $10n = (10 \times 50) = 500$

So, the number of employees in marketing department = $(800-500) = 300$

And, the number of employees in sales department = $(400-300) = 100$

The total number of employees = $(500+300+100) = 900$

Company B

The number of employees in HR department = $15n = (15 \times 50) = 750$

So, the number of employees in marketing department = $(1000-750) = 250$

And, the number of employees in sales department = $(600-250) = 350$

The total number of employees = $(750+250+350) = 1350$

Company C

The number of employees in HR department = $9n = (9 \times 50) = 450$

So, the number of employees in marketing department = $(600-450) = 150$

And, the number of employees in sales department = $(200-150) = 50$

The total number of employees = $(450+150+50) = 650$

Company D

The number of employees in HR department = $12n = (12 \times 50) = 600$

So, the number of employees in marketing department = $(1000-600) = 400$

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And, the number of employees in sales department = $(600-400) = 200$

The total number of employees = $(600+400+200) = 1200$

Company	The total number of employees	The number of employees in HR department	The number of employees in marketing department	The number of employees in sales department
A	900	500	300	100
B	1350	750	250	350
C	650	450	150	50
D	1200	600	400	200
Total	4100	2300	1100	700

Answer: A

Now, according to the question,

Company A

The total number of employees in marketing department = 300

The number of male employees in marketing department = 180

The number of female employees in marketing department = $(300-180) = 120$

The total number of employees in sales department = 100

The number of female employees in sales department = $(180/9) = 20$

The number of male employees in sales department = $(100-20) = 80$

Therefore, the required ratio = $120:80 = 3:2$.

Hence, the correct answer is option A.

7. Questions

Let the number of employees in marketing department of company A, B, C and D be $10n$, $15n$, $9n$ and $12n$ respectively.

Company A

The total number of employees in HR and marketing departments = $(400 \times 2) = 800$

The number of employees in HR department = $10n$

So, the number of employees in marketing department = $(800-10n)$

The total number of employees in marketing and sales departments = $(200 \times 2) = 400$

So, the number of employees in sales department of company = $\{400-(800-10n)\} = (10n-400)$

Company B

The total number of employees in HR and marketing departments = $(500 \times 2) = 1000$

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The number of employees in HR department = $15n$

So, the number of employees in marketing department = $(1000-15n)$

The total number of employees in marketing and sales departments = $(300 \times 2) = 600$

So, the number of employees in sales department of company = $\{600-(1000-15n)\} = (15n-400)$

Company C

The total number of employees in HR and marketing departments = $(300 \times 2) = 600$

The number of employees in HR department = $9n$

So, the number of employees in marketing department = $(600-9n)$

The total number of employees in marketing and sales departments = $(100 \times 2) = 200$

So, the number of employees in sales department of company = $\{200-(600-9n)\} = (9n-400)$

Company D

The total number of employees in HR and marketing departments = $(500 \times 2) = 1000$

The number of employees in HR department = $12n$

So, the number of employees in marketing department = $(1000-12n)$

The total number of employees in marketing and sales departments = $(300 \times 2) = 600$

So, the number of employees in sales department of company = $\{600-(1000-12n)\} = (12n-400)$

Now, the total number of employees in sales department of all four companies together =
 $\{(10n-400)+(15n-400)+(9n-400)+(12n-400)\} = (46n-1600)$

So,

$$46n - 1600 = 700$$

$$46n = 2300$$

$$n = 50$$

Company A

The number of employees in HR department = $10n = (10 \times 50) = 500$

So, the number of employees in marketing department = $(800-500) = 300$

And, the number of employees in sales department = $(400-300) = 100$

The total number of employees = $(500+300+100) = 900$

Company B

The number of employees in HR department = $15n = (15 \times 50) = 750$

So, the number of employees in marketing department = $(1000-750) = 250$

And, the number of employees in sales department = $(600-250) = 350$

The total number of employees = $(750+250+350) = 1350$

Company C

The number of employees in HR department = $9n = (9 \times 50) = 450$

So, the number of employees in marketing department = $(600-450) = 150$

And, the number of employees in sales department = $(200-150) = 50$

The total number of employees = $(450+150+50) = 650$

Company D

The number of employees in HR department = $12n = (12 \times 50) = 600$

So, the number of employees in marketing department = $(1000-600) = 400$

And, the number of employees in sales department = $(600-400) = 200$

The total number of employees = $(600+400+200) = 1200$

Company	The total number of employees	The number of employees in HR department	The number of employees in marketing department	The number of employees in sales department
A	900	500	300	100
B	1350	750	250	350
C	650	450	150	50
D	1200	600	400	200
Total	4100	2300	1100	700

Answer: D

Company B

The total number of employees in marketing department = 250

The number of graduate employees in marketing department = 150

So, the number of post-graduate employees in marketing department = $(250-150) = 100$.

The total number of employees in HR department = 750

The number of graduate employees in HR department = $\{(150/30) \times 100\} = 500$

The number of post-graduate employees in HR department = $(750-500) = 250$

Therefore, the required difference = $(250-100) = 150$.

Hence, the correct answer is option D.

8. Questions

Let the number of employees in marketing department of company A, B, C and D be $10n$, $15n$, $9n$ and $12n$ respectively.

Company A

The total number of employees in HR and marketing departments = $(400 \times 2) = 800$

The number of employees in HR department = $10n$

So, the number of employees in marketing department = $(800 - 10n)$

The total number of employees in marketing and sales departments = $(200 \times 2) = 400$

So, the number of employees in sales department of company = $\{400 - (800 - 10n)\} = (10n - 400)$

Company B

The total number of employees in HR and marketing departments = $(500 \times 2) = 1000$

The number of employees in HR department = $15n$

So, the number of employees in marketing department = $(1000 - 15n)$

The total number of employees in marketing and sales departments = $(300 \times 2) = 600$

So, the number of employees in sales department of company = $\{600 - (1000 - 15n)\} = (15n - 400)$

Company C

The total number of employees in HR and marketing departments = $(300 \times 2) = 600$

The number of employees in HR department = $9n$

So, the number of employees in marketing department = $(600 - 9n)$

The total number of employees in marketing and sales departments = $(100 \times 2) = 200$

So, the number of employees in sales department of company = $\{200 - (600 - 9n)\} = (9n - 400)$

Company D

The total number of employees in HR and marketing departments = $(500 \times 2) = 1000$

The number of employees in HR department = $12n$

So, the number of employees in marketing department = $(1000 - 12n)$

The total number of employees in marketing and sales departments = $(300 \times 2) = 600$

So, the number of employees in sales department of company = $\{600 - (1000 - 12n)\} = (12n - 400)$

Now, the total number of employees in sales department of all four companies together =
 $\{(10n - 400) + (15n - 400) + (9n - 400) + (12n - 400)\} = (46n - 1600)$

So,

$$46n - 1600 = 700$$

$$46n = 2300$$

$$n = 50$$

Company A

The number of employees in HR department = $10n = (10 \times 50) = 500$

So, the number of employees in marketing department = $(800 - 500) = 300$

And, the number of employees in sales department = $(400 - 300) = 100$

The total number of employees = $(500 + 300 + 100) = 900$

Company B

The number of employees in HR department = $15n = (15 \times 50) = 750$

So, the number of employees in marketing department = $(1000 - 750) = 250$

And, the number of employees in sales department = $(600 - 250) = 350$

The total number of employees = $(750 + 250 + 350) = 1350$

Company C

The number of employees in HR department = $9n = (9 \times 50) = 450$

So, the number of employees in marketing department = $(600 - 450) = 150$

And, the number of employees in sales department = $(200 - 150) = 50$

The total number of employees = $(450 + 150 + 50) = 650$

Company D

The number of employees in HR department = $12n = (12 \times 50) = 600$

So, the number of employees in marketing department = $(1000 - 600) = 400$

And, the number of employees in sales department = $(600 - 400) = 200$

The total number of employees = $(600 + 400 + 200) = 1200$

Company	The total number of employees	The number of employees in HR department	The number of employees in marketing department	The number of employees in sales department
A	900	500	300	100
B	1350	750	250	350
C	650	450	150	50
D	1200	600	400	200
Total	4100	2300	1100	700

Answer: E

I: The ratio of the number of employees in HR and sales department of company C = $450:50 = 9:1$.

So, I is definitely not true.

II: The difference between the number of employees in marketing department of company C and D = $(400 - 150) = 250$.

So, II is definitely not true.

III: The total number of employees in company B = 1350

So, III is definitely not true.

Therefore, none is true.

Hence, the correct answer is option E.

9. Questions

Let the number of employees in marketing department of company A, B, C and D be $10n$, $15n$, $9n$ and $12n$ respectively.

Company A

The total number of employees in HR and marketing departments = $(400 \times 2) = 800$

The number of employees in HR department = $10n$

So, the number of employees in marketing department = $(800 - 10n)$

The total number of employees in marketing and sales departments = $(200 \times 2) = 400$

So, the number of employees in sales department of company = $\{400 - (800 - 10n)\} = (10n - 400)$

Company B

The total number of employees in HR and marketing departments = $(500 \times 2) = 1000$

The number of employees in HR department = $15n$

So, the number of employees in marketing department = $(1000 - 15n)$

The total number of employees in marketing and sales departments = $(300 \times 2) = 600$

So, the number of employees in sales department of company = $\{600 - (1000 - 15n)\} = (15n - 400)$

Company C

The total number of employees in HR and marketing departments = $(300 \times 2) = 600$

The number of employees in HR department = $9n$

So, the number of employees in marketing department = $(600 - 9n)$

The total number of employees in marketing and sales departments = $(100 \times 2) = 200$

So, the number of employees in sales department of company = $\{200 - (600 - 9n)\} = (9n - 400)$

Company D

The total number of employees in HR and marketing departments = $(500 \times 2) = 1000$

The number of employees in HR department = $12n$

So, the number of employees in marketing department = $(1000 - 12n)$

The total number of employees in marketing and sales departments = $(300 \times 2) = 600$

So, the number of employees in sales department of company = $\{600-(1000-12n)\} = (12n-400)$

Now, the total number of employees in sales department of all four companies together =
 $\{(10n-400)+(15n-400)+(9n-400)+(12n-400)\} = (46n-1600)$

So,

$$46n - 1600 = 700$$

$$46n = 2300$$

$$n = 50$$

Company A

The number of employees in HR department = $10n = (10 \times 50) = 500$

So, the number of employees in marketing department = $(800-500) = 300$

And, the number of employees in sales department = $(400-300) = 100$

The total number of employees = $(500+300+100) = 900$

Company B

The number of employees in HR department = $15n = (15 \times 50) = 750$

So, the number of employees in marketing department = $(1000-750) = 250$

And, the number of employees in sales department = $(600-250) = 350$

The total number of employees = $(750+250+350) = 1350$

Company C

The number of employees in HR department = $9n = (9 \times 50) = 450$

So, the number of employees in marketing department = $(600-450) = 150$

And, the number of employees in sales department = $(200-150) = 50$

The total number of employees = $(450+150+50) = 650$

Company D

The number of employees in HR department = $12n = (12 \times 50) = 600$

So, the number of employees in marketing department = $(1000-600) = 400$

And, the number of employees in sales department = $(600-400) = 200$

The total number of employees = $(600+400+200) = 1200$

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Company	The total number of employees	The number of employees in HR department	The number of employees in marketing department	The number of employees in sales department
A	900	500	300	100
B	1350	750	250	350
C	650	450	150	50
D	1200	600	400	200
Total	4100	2300	1100	700

Answer: C

Company F

The total number of employees working = 800

The number of employees working in sales department = $(200/2) = 100$

Therefore, the number of employees working in marketing department of company F = $\{3/7 \text{ of } (800-100)\} = 3/7 \text{ of } 700 = 300$.

Hence, the correct answer is option C.

10. Questions

Let the number of employees in marketing department of company A, B, C and D be $10n$, $15n$, $9n$ and $12n$ respectively.

Company A

The total number of employees in HR and marketing departments = $(400 \times 2) = 800$

The number of employees in HR department = $10n$

So, the number of employees in marketing department = $(800-10n)$

The total number of employees in marketing and sales departments = $(200 \times 2) = 400$

So, the number of employees in sales department of company = $\{400-(800-10n)\} = (10n-400)$

Company B

The total number of employees in HR and marketing departments = $(500 \times 2) = 1000$

The number of employees in HR department = $15n$

So, the number of employees in marketing department = $(1000-15n)$

The total number of employees in marketing and sales departments = $(300 \times 2) = 600$

So, the number of employees in sales department of company = $\{600-(1000-15n)\} = (15n-400)$

Company C

The total number of employees in HR and marketing departments = $(300 \times 2) = 600$

The number of employees in HR department = $9n$

So, the number of employees in marketing department = $(600-9n)$

The total number of employees in marketing and sales departments = $(100 \times 2) = 200$

So, the number of employees in sales department of company = $\{200-(600-9n)\} = (9n-400)$

Company D

The total number of employees in HR and marketing departments = $(500 \times 2) = 1000$

The number of employees in HR department = $12n$

So, the number of employees in marketing department = $(1000-12n)$

The total number of employees in marketing and sales departments = $(300 \times 2) = 600$

So, the number of employees in sales department of company = $\{600-(1000-12n)\} = (12n-400)$

Now, the total number of employees in sales department of all four companies together =
 $\{(10n-400)+(15n-400)+(9n-400)+(12n-400)\} = (46n-1600)$

So,

$$46n - 1600 = 700$$

$$46n = 2300$$

$$n = 50$$

Company A

The number of employees in HR department = $10n = (10 \times 50) = 500$

So, the number of employees in marketing department = $(800-500) = 300$

And, the number of employees in sales department = $(400-300) = 100$

The total number of employees = $(500+300+100) = 900$

Company B

The number of employees in HR department = $15n = (15 \times 50) = 750$

So, the number of employees in marketing department = $(1000-750) = 250$

And, the number of employees in sales department = $(600-250) = 350$

The total number of employees = $(750+250+350) = 1350$

Company C

The number of employees in HR department = $9n = (9 \times 50) = 450$

So, the number of employees in marketing department = $(600-450) = 150$

And, the number of employees in sales department = $(200-150) = 50$

The total number of employees = $(450+150+50) = 650$

Company D

The number of employees in HR department = $12n = (12 \times 50) = 600$

So, the number of employees in marketing department = $(1000 - 600) = 400$

And, the number of employees in sales department = $(600 - 400) = 200$

The total number of employees = $(600 + 400 + 200) = 1200$

Company	The total number of employees	The number of employees in HR department	The number of employees in marketing department	The number of employees in sales department
A	900	500	300	100
B	1350	750	250	350
C	650	450	150	50
D	1200	600	400	200
Total	4100	2300	1100	700

Answer: E

X = The difference between the number of employees working in marketing and sales department of company B = $(350 - 250) = 100$

Y = The number of employees working in sales departments of company A = 100

Therefore, $X = Y$.

Hence, the correct answer is option E.

11. Questions

Answer: A

Series I: 11, 15, 8, 18, __[P]__, 21

1st term = 11

2nd term = $(11 + 4) = 15$

3rd term = $(15 - 7) = 8$

4th term = $(8 + 10) = 18$

5th term = $(18 - 13) = 5$ (P)

6th term = $(5 + 16) = 21$

So, the missing term = $P = 5$

Series II: 5, 7, 17, 55, 225, __[Q]__

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$$1^{\text{st}} \text{ term} = 5$$

$$2^{\text{nd}} \text{ term} = (5 \times 1 + 2) = (5 + 2) = 7$$

$$3^{\text{rd}} \text{ term} = (7 \times 2 + 3) = (14 + 3) = 17$$

$$4^{\text{th}} \text{ term} = (17 \times 3 + 4) = (51 + 4) = 55$$

$$5^{\text{th}} \text{ term} = (55 \times 4 + 5) = (220 + 5) = 225$$

$$6^{\text{th}} \text{ term} = (225 \times 5 + 6) = (1125 + 6) = 1131 \text{ (Q)}$$

So, the missing term = Q = 1131

I: Only 5 is exactly divisible by 5.

So, I is definitely not true.

II: 1131 is a four-digit natural number.

So, II is definitely true.

III: 5 is not an odd composite number.

So, III is definitely not true.

Therefore, only II is definitely true.

Hence, the correct answer is option A.

12. Questions

Answer: D

Series I: 8, 32, 4, 12, 2, 8

$$1^{\text{st}} \text{ term} = 8$$

$$2^{\text{nd}} \text{ term} = (8 \times 4) = 32$$

$$3^{\text{rd}} \text{ term} = (32/8) = 4$$

$$4^{\text{th}} \text{ term} = (4 \times 4) = 16 \text{ (12 is wrong)}$$

$$5^{\text{th}} \text{ term} = (16/8) = 2$$

$$6^{\text{th}} \text{ term} = (2 \times 4) = 8$$

So, the wrong term = 12

And, the correct term = 16

$$4^2 = 16$$

$$2^4 = 16$$

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Hence, the correct answer is option D.

13. Questions

Answer: B

Series: 3, 19, 55, 119, 219, __[?]

$$1^{\text{st}} \text{ term} = 3$$

$$2^{\text{nd}} \text{ term} = (3+4^2) = (3+16) = 19$$

$$3^{\text{rd}} \text{ term} = (19+6^2) = (19+36) = 55$$

$$4^{\text{th}} \text{ term} = (55+8^2) = (55+64) = 119$$

$$5^{\text{th}} \text{ term} = (119+10^2) = (119+100) = 219$$

$$6^{\text{th}} \text{ term} = (219+12^2) = (219+144) = 363 (?)$$

So, the missing term = ? = 363

Hence, the correct answer is option B.

14. Questions

Answer: A

Series I: 40, 48, 64, 92, 160, 288

$$1^{\text{st}} \text{ term} = 40$$

$$2^{\text{nd}} \text{ term} = (40+8) = 48$$

$$3^{\text{rd}} \text{ term} = (48+16) = 64$$

$$4^{\text{th}} \text{ term} = (64+32) = 96 \text{ (92 is wrong)}$$

$$5^{\text{th}} \text{ term} = (96+64) = 160$$

$$6^{\text{th}} \text{ term} = (160+128) = 288$$

So, the wrong term = 92

And, the correct term = 96

Series II: 20, 28, 44, 76, 140, 260

$$1^{\text{st}} \text{ term} = 20$$

$$2^{\text{nd}} \text{ term} = (20+8) = 28$$

$$3^{\text{rd}} \text{ term} = (28+16) = 44$$

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$$4^{\text{th}} \text{ term} = (44+32) = 76$$

$$5^{\text{th}} \text{ term} = (76+64) = 140$$

$$6^{\text{th}} \text{ term} = (140+128) = 268 \text{ (260 is wrong)}$$

So, the wrong term = 260

And, the correct term = 268

Therefore, the required difference = $(260-92) = 168$.

Hence, the correct answer is option A.

15. Questions

Answer: C

Series: 12, 13, 5, 32, __[?], 93

$$1^{\text{st}} \text{ term} = 12$$

$$2^{\text{nd}} \text{ term} = (12+1^3) = (12+1) = 13$$

$$3^{\text{rd}} \text{ term} = (13-2^3) = (13-8) = 5$$

$$4^{\text{th}} \text{ term} = (5+3^3) = (5+27) = 32$$

$$5^{\text{th}} \text{ term} = (32-4^3) = (32-64) = -32 \text{ (?)}$$

$$6^{\text{th}} \text{ term} = (-32+5^3) = (-32+125) = 93$$

So, the missing term = ? = -32

Hence, the correct answer is option C.

16. Questions

City P

Let the number of people who have only android and only iPhone be $2m$ and m respectively.

The number of people who have both android and iPhone = 100

So, the total number of people who have android = $(2m+100)$

And, the total number of people who have iPhone = $(m+100)$

Now,

$$(2m+100)/(m+100) = 100/60$$

$$(2m+100)/(m+100) = 5/3$$

$$6m + 300 = 5m + 500$$

$$m = 200$$

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Therefore,

The number of people who have only android = $2m = (2 \times 200) = 400$

The number of people who have only iPhone = $m = 200$

The number of people who have both android and iPhone = 100

So, the total number of people who have mobiles = $(400+200+100) = 700$

City Q

The ratio of the number of people who have only android and only iPhone = $4:3$. Let it be $4n$ and $3n$ respectively.

The number of people who have both android and iPhone = 60

So, the total number of people who have android = $(4n+60)$

And, the total number of people who have iPhone = $(3n+60)$

Now,

$$(4n+60)/(3n+60) = 100/80$$

$$(4n+60)/(3n+60) = 5/4$$

$$16n + 240 = 15n + 300$$

$$n = 60$$

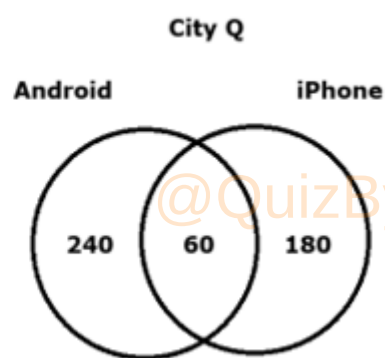
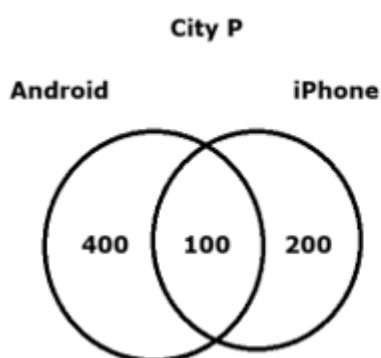
Therefore,

The number of people who have only android = $4n = (4 \times 60) = 240$

The number of people who have only iPhone = $3n = (3 \times 60) = 180$

The number of people who have both android and iPhone = 60

So, the total number of people who have mobiles = $(240+180+60) = 480$



Answer: B

Now, according to the question,

City P

The total number of people who have android = $(400+100) = 500$

The total number of people who have iPhone = $(200+100) = 300$

The ratio of the total number of people who have 5G android and iPhone = $150:100 = 3:2$. Let it be $3k$ and $2k$ respectively.

So, the total number of people who have 4G android = $(500-3k)$

And, the total number of people who have 4G iPhone = $(300-2k)$

Now,

$$(500-3k) - (300-2k) = 150$$

$$350 - 3k = 300 - 2k$$

$$k = 50$$

Therefore, the total number of people who have 5G mobiles (android + iPhone) in city P = $(3k+2k) = 5k = (5 \times 50) = 250$.

Hence, the correct answer is option B.

17. Questions

City P

Let the number of people who have only android and only iPhone be $2m$ and m respectively.

The number of people who have both android and iPhone = 100

So, the total number of people who have android = $(2m+100)$

And, the total number of people who have iPhone = $(m+100)$

Now,

$$(2m+100)/(m+100) = 100/60$$

$$(2m+100)/(m+100) = 5/3$$

$$6m + 300 = 5m + 500$$

$$m = 200$$

Therefore,

The number of people who have only android = $2m = (2 \times 200) = 400$

The number of people who have only iPhone = $m = 200$

The number of people who have both android and iPhone = 100

So, the total number of people who have mobiles = $(400+200+100) = 700$

City Q

The ratio of the number of people who have only android and only iPhone = $4:3$. Let it be $4n$ and $3n$ respectively.

The number of people who have both android and iPhone = 60

So, the total number of people who have android = $(4n+60)$

And, the total number of people who have iPhone = $(3n+60)$

Now,

$$(4n+60)/(3n+60) = 100/80$$

$$(4n+60)/(3n+60) = 5/4$$

$$16n + 240 = 15n + 300$$

$$n = 60$$

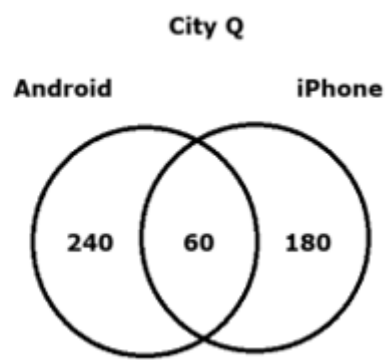
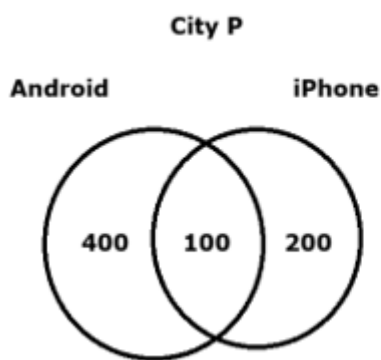
Therefore,

The number of people who have only android = $4n = (4 \times 60) = 240$

The number of people who have only iPhone = $3n = (3 \times 60) = 180$

The number of people who have both android and iPhone = 60

So, the total number of people who have mobiles = $(240+180+60) = 480$



Answer: A

City Q

The total number of people who have android = $(240+60) = 300$

The total number of people who have iPhone = $(180+60) = 240$

The total number of males who have android = 175

So, the total number of males who have iPhone = $(175-15) = 160$ @QuizByMickey

Now, the total number of females who have android = $(300-175) = 125$

And, the total number of females who iPhone = $(240-160) = 80$

Therefore, the required ratio = $125:80 = 25:16$.

Hence, the correct answer is option A.

18. Questions

City P

Let the number of people who have only android and only iPhone be $2m$ and m respectively.

The number of people who have both android and iPhone = 100

So, the total number of people who have android = $(2m+100)$

And, the total number of people who have iPhone = $(m+100)$

Now,

$$(2m+100)/(m+100) = 100/60$$

$$(2m+100)/(m+100) = 5/3$$

$$6m + 300 = 5m + 500$$

$$m = 200$$

Therefore,

The number of people who have only android = $2m = (2 \times 200) = 400$

The number of people who have only iPhone = $m = 200$

The number of people who have both android and iPhone = 100

So, the total number of people who have mobiles = $(400+200+100) = 700$

City Q

The ratio of the number of people who have only android and only iPhone = 4:3. Let it be $4n$ and $3n$ respectively.

The number of people who have both android and iPhone = 60

So, the total number of people who have android = $(4n+60)$

And, the total number of people who have iPhone = $(3n+60)$

Now,

$$(4n+60)/(3n+60) = 100/80$$

$$(4n+60)/(3n+60) = 5/4$$

$$16n + 240 = 15n + 300$$

$$n = 60$$

Therefore,

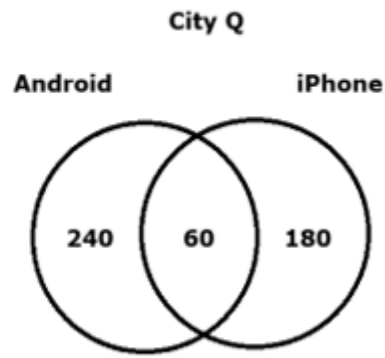
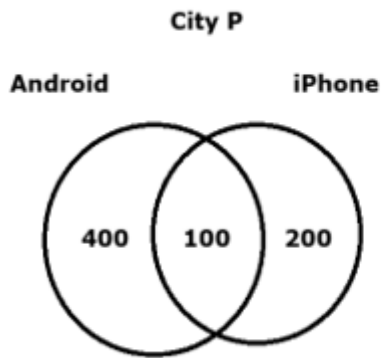
@QuizByMickey

The number of people who have only android = $4n = (4 \times 60) = 240$

The number of people who have only iPhone = $3n = (3 \times 60) = 180$

The number of people who have both android and iPhone = 60

So, the total number of people who have mobiles = $(240+180+60) = 480$



Answer: D

City R

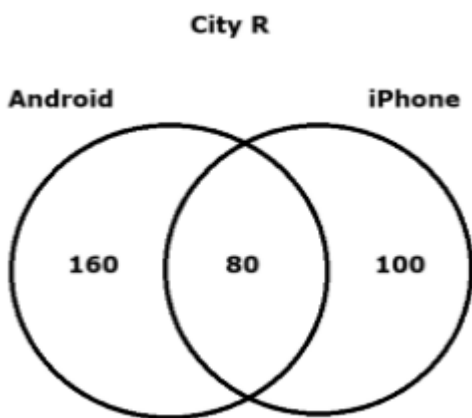
The total number of people who have mobiles = $(480 - 140) = 340$

The number of people who have both android and iPhone = 80

So, the number of people who have only android = $\frac{8}{13}$ of $(340 - 80) = \frac{8}{13}$ of 260 = 160

Therefore, the total number of who have android in city R = $(160 + 80) = 240$.

Hence, the correct answer is option D.



19. Questions

City P

Let the number of people who have only android and only iPhone be $2m$ and m respectively.

The number of people who have both android and iPhone = 100 @QuizByMickey

So, the total number of people who have android = $(2m + 100)$

And, the total number of people who have iPhone = $(m + 100)$

Now,

$$(2m + 100) / (m + 100) = 100 / 60$$

$$(2m + 100) / (m + 100) = 5 / 3$$

$$6m + 300 = 5m + 500$$

$$m = 200$$

Therefore,

$$\text{The number of people who have only android} = 2m = (2 \times 200) = 400$$

$$\text{The number of people who have only iPhone} = m = 200$$

$$\text{The number of people who have both android and iPhone} = 100$$

$$\text{So, the total number of people who have mobiles} = (400 + 200 + 100) = 700$$

City Q

The ratio of the number of people who have only android and only iPhone = 4:3. Let it be $4n$ and $3n$ respectively.

$$\text{The number of people who have both android and iPhone} = 60$$

$$\text{So, the total number of people who have android} = (4n + 60)$$

$$\text{And, the total number of people who have iPhone} = (3n + 60)$$

Now,

$$(4n + 60) / (3n + 60) = 100 / 80$$

$$(4n + 60) / (3n + 60) = 5 / 4$$

$$16n + 240 = 15n + 300$$

$$n = 60$$

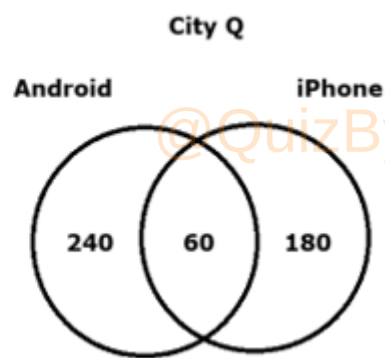
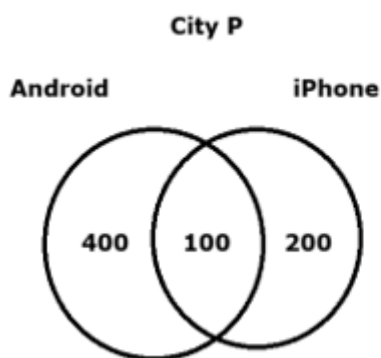
Therefore,

$$\text{The number of people who have only android} = 4n = (4 \times 60) = 240$$

$$\text{The number of people who have only iPhone} = 3n = (3 \times 60) = 180$$

$$\text{The number of people who have both android and iPhone} = 60$$

$$\text{So, the total number of people who have mobiles} = (240 + 180 + 60) = 480$$



Answer: B

City P

$$\text{The total number of people who have symbian} = 72\% \text{ of } (400 + 100) = 72\% \text{ of } 500 = 360.$$

The total number of people who have iPhone = $(200+100) = 300$

Therefore, the required difference = $(360-300) = 60$.

Hence, the correct answer is option B.

20. Questions

City P

Let the number of people who have only android and only iPhone be $2m$ and m respectively.

The number of people who have both android and iPhone = 100

So, the total number of people who have android = $(2m+100)$

And, the total number of people who have iPhone = $(m+100)$

Now,

$$(2m+100)/(m+100) = 100/60$$

$$(2m+100)/(m+100) = 5/3$$

$$6m + 300 = 5m + 500$$

$$m = 200$$

Therefore,

The number of people who have only android = $2m = (2 \times 200) = 400$

The number of people who have only iPhone = $m = 200$

The number of people who have both android and iPhone = 100

So, the total number of people who have mobiles = $(400+200+100) = 700$

City Q

The ratio of the number of people who have only android and only iPhone = 4:3. Let it be $4n$ and $3n$ respectively.

The number of people who have both android and iPhone = 60

So, the total number of people who have android = $(4n+60)$

And, the total number of people who have iPhone = $(3n+60)$

Now,

$$(4n+60)/(3n+60) = 100/80$$

$$(4n+60)/(3n+60) = 5/4$$

$$16n + 240 = 15n + 300$$

$$n = 60$$

Therefore,

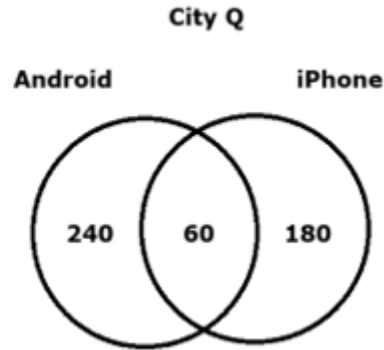
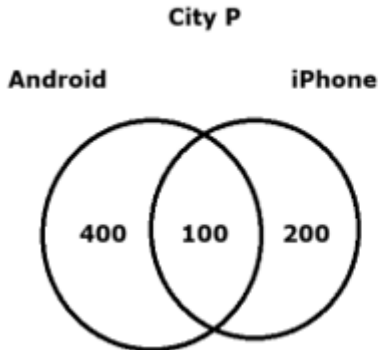
@QuizByMickey

The number of people who have only android = $4n = (4 \times 60) = 240$

The number of people who have only iPhone = $3n = (3 \times 60) = 180$

The number of people who have both android and iPhone = 60

So, the total number of people who have mobiles = $(240 + 180 + 60) = 480$



Answer: A

City Q

The number of people who have only android = 240

The number of people who have only iPhone = 180

The increased number of people who have both android and iPhone = $(60 + X)$

Now,

$$240 + 180 + (60 + X) = 500$$

$$X = 500 - 480$$

$$X = 20$$

Therefore, the value of $(\sqrt{X-4}) = \sqrt{(20-4)} = \sqrt{16} = 4$.

Hence, the correct answer is option A.

@QuizByMickey